

Total No. of Questions : 8]

SEAT No. :

**PD4590**

[Total No. of Pages : 2

**[6404]-95**

**B.E (Computer Engineering)**

**DEEP LEARNING**

**(2019 Pattern) (410251) (Semester - VIII)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Make suitable assumption whenever necessary.*

**Q1)** a) Draw and Explain CNN Architecture in detail. [6]

b) Explain working of Convolution Layer. [6]

c) Explain pooling layers and its type. [6]

OR

**Q2)** a) Explain all the features of pooling layer. [6]

b) Explain local response Normalization. [6]

c) Explain ReLU Layer in detail. [6]

**Q3)** a) Explain recursive neural network. [6]

b) Explain the LSTM in RNN. [6]

c) Explain in brief about working of RNN. [5]

OR

**Q4)** a) Difference between CNN Vs RNN. [6]

b) What are the challenges of long term dependencies? [6]

c) Explain Encoder-Decoder RNN model. [5]

**P.T.O.**

- Q5)** a) Explain Deep Generative Model. [6]  
b) Explain Boltzmann Machine in details. [6]  
c) Explain in brief GAN with an example. [6]

OR

- Q6)** a) Explain deep belief networks in detail. [6]  
b) What is Generative Adversarial Network? Explain its component. [6]  
c) Explain types of GAN. [6]

- Q7)** a) Explain Markov Decision Process. [6]  
b) Explain deep reinforcement learning. [6]  
c) What are the challenges of reinforcement learning? [5]

OR

- Q8)** a) Explain the process of Deep Q-learning. [6]  
b) Explain reinforcement learning for Tic-Tac-Toe game. [6]  
c) Explain Dynamic programming algorithm for reinforcement learning. [5]

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